Is Economic Growth Decoupling from CO_2 Emissions?

A cross-country analysis of the Our World in Data CO_2 dataset, 2005–2022

Your Name (Student ID)
MAT2007 Introduction to Programming, Maastricht University

October 2026

Introduction

Limiting global warming requires deep cuts in CO_2 emissions, while most countries still pursue economic growth. Whether the two goals are compatible depends on *decoupling*. When CO_2 emissions grow more slowly than GDP, we have *relative* decoupling. When emissions fall while GDP keeps growing, we have *absolute* decoupling [2]. A common objection to apparent decoupling in rich countries is that they have only moved polluting production abroad. In this report we ask: **how many countries have achieved absolute decoupling since 2005? How does this depend on a country's income level? And does the result survive when emissions are counted by consumption rather than production?**

Dataset

We use the CO_2 and Greenhouse Gas Emissions dataset published by Our World in Data (OWID) [1]. It is a single CSV file of about 14 MB with 79 columns and 50 411 rows, one per country (or region) per year. We use these columns:

- `gdp`: GDP in constant international dollars (Maddison Project), available up to 2022;
- `co2`: annual territorial (production-based) fossil CO_2 emissions in Mt;
- `consumption_co2`: emissions adjusted for trade, i.e. attributed to the country where the goods are consumed;
- `population`, `country`, `iso_code`, `year`.

We clean the data in three steps. First, we remove regional aggregates (rows with no ISO code or with an `OWID_` code). Second, we keep only countries with complete GDP, CO_2 and population data for every year from 2005 to 2022. Third, we drop countries with fewer than one million inhabitants. This leaves $N = 153$ countries. Consumption-based data exist for 116 of them.

Method

The analysis is implemented in Python (`pandas`, `scipy`, `matplotlib`). For each country we fit a log-linear trend $\ln X(t) = a + b\,t$ by least squares, separately for $X =$ GDP, territorial CO_2 and consumption CO_2. The average annual growth rate is $r = 100\,(e^{b} - 1)\,\%$. Its uncertainty, propagated from the standard error of the slope, is $\sigma_r = 100\,e^{b}\,\sigma_b$. Fitting a trend uses all 18 years of data, so it is less sensitive to one-off years such as 2020 than a simple start-to-end comparison would be. Each country is then classified:

- *absolute decoupling*: $r_{\mathrm{GDP}} > 0$ and CO_2 decreasing at 95% confidence, i.e. $r_{\mathrm{CO_2}} + 1.96\,\sigma_r < 0$;
- *relative decoupling*: $0 < r_{\mathrm{GDP}}$ and $r_{\mathrm{CO_2}} < r_{\mathrm{GDP}}$ (and not absolute);
- *no decoupling*: $r_{\mathrm{CO_2}} \geq r_{\mathrm{GDP}} > 0$;
- *GDP decline*: $r_{\mathrm{GDP}} \leq 0$. Decoupling is not defined here (e.g. Greece, Italy, Venezuela).

Countries are divided into four income groups using the quartiles of their GDP per capita in 2005. For the fraction of countries $p = k/n$ in a class we quote the binomial uncertainty $\sigma_p = \sqrt{p(1-p)/n}$.

Results and discussion

Figure 1 shows the GDP growth rate against the territorial CO_2 growth rate for all 153 countries. Using territorial emissions, the countries split as follows: $(22.2 \pm 3.4)\%$ absolute decoupling (34 countries), $(43.8 \pm 4.0)\%$ relative decoupling, $(26.8 \pm 3.6)\%$ no decoupling, and $(7.2 \pm 2.1)\%$ GDP decline. Absolute decoupling is concentrated in rich

economies. For example, the United Kingdom grew by (1.11±0.16)%/yr while its emissions fell by (3.54±0.15)%/yr. For the United States the corresponding figures are (1.68 ± 0.10)%/yr and $-(1.22 \pm 0.14)$%/yr, and for Germany (1.58 ± 0.12)%/yr and $-(1.53 \pm 0.19)$%/yr. China ($r_{\mathrm{GDP}} = 7.05\%$, $r_{\mathrm{CO_2}} = 3.51\%$) and India (6.27%, 5.05%) show only relative decoupling. The initial income level and the CO_2 growth rate are strongly anti-correlated (Spearman $\rho = -0.69$, $p \approx 10^{-22}$).

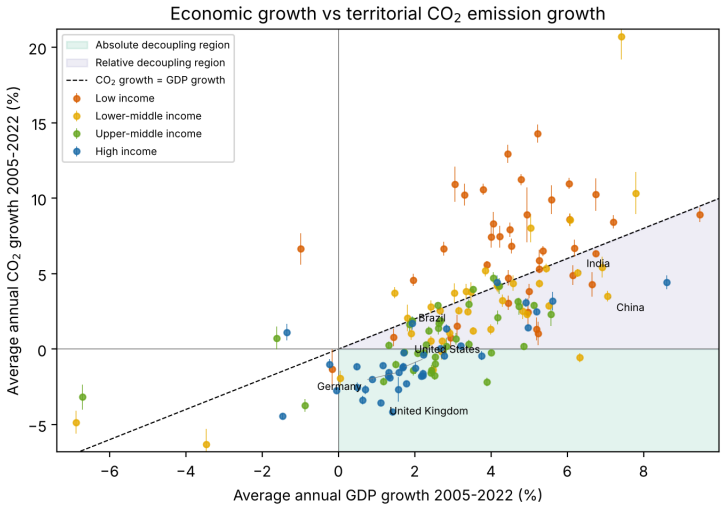


Figure 1: Average annual GDP growth versus territorial CO_2 growth, 2005–2022. Each point is a country, coloured by income quartile in 2005. The error bars are the 1σ uncertainties of the fitted CO_2 trend. Points in the green region show absolute decoupling. Points in the purple region show relative decoupling.

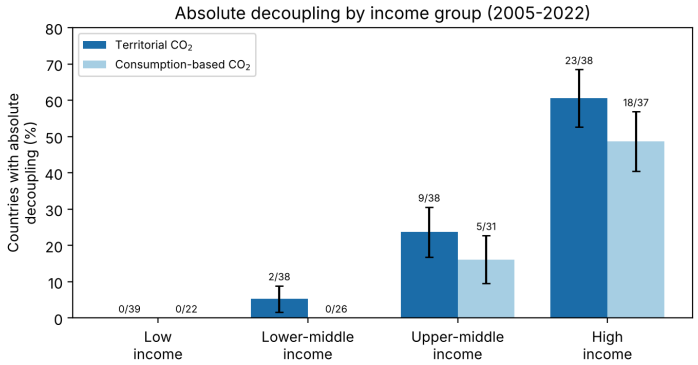


Figure 2: Fraction of countries with absolute decoupling in each income quartile, for territorial and consumption-based emissions. The error bars are binomial uncertainties. The labels above the bars give the raw counts k/n.

Figure 2 quantifies the income dependence. No low-income country (0/39) and only 2/38 lower-middle-income countries decoupled absolutely, against $(60.5 \pm 7.9)\%$ (23/38) of high-income countries. The difference between the high- and low-income groups is 0.61 ± 0.08, which is 7.6σ. Counting emissions by consumption lowers the high-income fraction to $(48.6 \pm 8.2)\%$ (18/37), a reduction of 1.0σ. Offshoring of production therefore explains part of the apparent decoupling, but not most of it. For example, the UK, the US, Germany and the Netherlands all still show absolute decoupling on a consumption basis.

Conclusion. Absolute decoupling of GDP from CO_2 emissions is real but limited. It occurred in about one country in five over 2005–2022, almost exclusively in high- and upper-middle-income economies, and it largely persists when trade is taken into account. However, even the decoupling countries reduce emissions by only about 1–4%/yr. This is well below the roughly 6%/yr global cut implied by 1.5 °C pathways (CO_2 about 48% below 2019 levels by 2030) [2]. Most of the world, including the large emitters China and India, is still in the relative-decoupling regime. One limitation is that the growth rates assume a constant exponential trend, which ignores changes in trend within the period.

References

[1] Our World in Data, *CO_2 and Greenhouse Gas Emissions dataset*, `https://github.com/owid/co2-data` (commit `382ee6c`, accessed October 2026).

[2] IPCC, *Climate Change 2022: Mitigation of Climate Change*, Working Group III contribution to the Sixth Assessment Report, Cambridge University Press (2022).